

Python 调用海思 API 说明



版本历史

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前言

Ubuntu 系统下使用 Python 调用海思 API，本文介绍使用 pybind11 的方式；从 VPSS 获取视频帧并使用 python opencv 处理后发送给 VO 显示。

1. 板端环境部署

烧录 Ubuntu 固件后，安装软件

```
#安装 python
apt install python3.10-dev
#apt remove python3.10-dev

#安装 pip3
apt install python3-pip -y

#板端安装
#apt install python3-opencv
apt remove python3-opencv

#安装无 GUI 版
pip install opencv-python-headless -i https://pypi.tuna.tsinghua.edu.cn/simple

#建议安装无 GUI 版本的 opencv，python 在 import 速度相对快一些
```

```
root@localhost:~# python3 -V
Python 3.10.12
root@localhost:~#
```

执行下面命令显示版本号，说明安装成功。

```
python3 -c "import cv2; print(cv2.__version__)"
```

2. 程序移植

(1) 拷贝板端 python 头文件至 mpp 下的 src/pymodule/include

```
/usr/include/python3.10
/usr/include/aarch64-linux-gnu/python3.10
```

将 两 个 文 件 下 放 到 src/pymodule/include 下 ； 也 可 以 用

/usr/include/aarch64-linux-gnu/python3.10/pyconfig.h 覆盖/usr/include/python3.10/pyconfig.h。

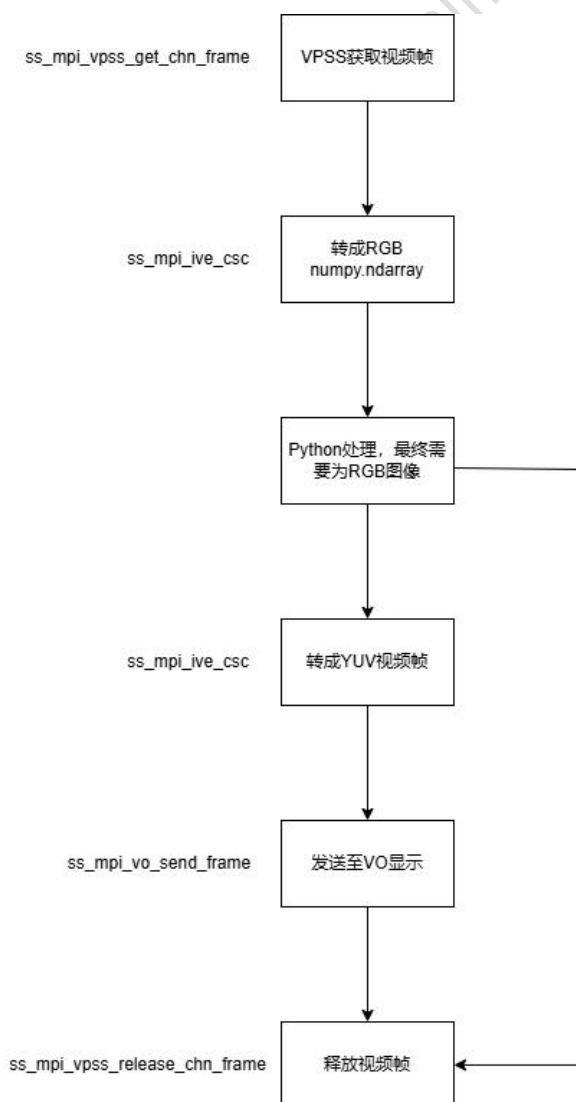
(2) 下载 pybind11 头文件

```
ebaina@Ubuntu22: ~/Workspace/opensource
ebaina@Ubuntu22:~/Workspace/opensource$ git clone https://github.com/pybind/pybind11.git
Cloning into 'pybind11'...
remote: Enumerating objects: 28338, done.
remote: Total 28338 (delta 0), reused 0 (delta 0), pack-reused 28338 (from 1)
Receiving objects: 100% (28338/28338), 12.43 MiB | 116.00 KiB/s, done.
Resolving deltas: 100% (20060/20060), done.
ebaina@Ubuntu22:~/Workspace/opensource$
```

拷贝该项目下的 pybind11/include/pybind11 至 src/pymodule/include。

```
ebaina@Ubuntu22:~/Workspace/SS928V100_SDK_V2.0.2.2_MPP_Sample/src/pymodule$ ls include/pybind11/
attr.h      complex.h    eigen.h      gil_safe_call_once.h  operators.h  stl_bind.h      typing.h
buffer_info.h  conduit      embed.h      gil_simple.h          options.h    stl.h           warnings.h
cast.h        critical_section.h  eval.h       iostream.h           pybind11.h  subinterpreter.h
chrono.h       detail        functional.h  native_enum.h        pytypes.h   trampoline_self_life_support.h
common.h       eigen         gil.h        numpy.h              stl         type_caster_pyobject_ptr.h
ebaina@Ubuntu22:~/Workspace/SS928V100_SDK_V2.0.2.2_MPP_Sample/src/pymodule$
```

(3) VPSS 获取视频帧并使用 python opencv 处理后发送给 VO 显示



(4) 编译

```
eabina@ubuntu22:~/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/src/pymodule$ tree -L 2
```

```
tree
├── .
│   ├── pymod1
│   └── thm3.10
├── Makefile
└── sample_pymodule.cpp
```

```
eabina@ubuntu22:~/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/src/pymodule$ make
```

```
aarch64-linux-gnu-gcc -DS5928V100 -DOR_XXX -lpthread -ln -ld -DISP V2 -lstdc++ -Wall -fsigned-char -mcpu=cortex-a53 -fno-aggressive-loop-optimizations -ld -ffunction-sections -fdats-sections -O2 -fstack-protector-strong -fpic -D_FILE_OFFSET_BITS=64 -DLARGEFILE_SOURCE -D_LARGEFILE64_SOURCE -PIE -pie -s -Wl,-flang=chdr -DOT_RELAX -OVER_XU1 -OVER_YuB -OVER_ZuB -OVER_B=10 -OVERS_BIT_64 -DKERNEL_BIT_64 -Wno-date-time -DENOSDR0_TYPE=SONY_IMX374_SLAVE_MIPI_A0_30FPS_12BIT -DENOSDR1_TYPE=SONY_IMX374_SLAVE_MIPI_A0_30FPS_12BIT -DENOSDR2_TYPE=HLSI_CS1860_MIPI_A0_30FPS_10BIT -DENOSDR3_TYPE=HLSI_CS1860_MIPI_A0_30FPS_10BIT -DOT_ACUTE_TYPE_HMT -DOT_ACUTE_TYPE_INNER -DOT_USE_STATIC_MODULE_REGISTER -DOT_MAC_USE_STATIC_MODULE_REGISTER -DOT_MAC_HAVE_SBR_LN -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/src/common -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/histctrl/mip -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/histctrl/inc/exp -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/histctrl/osal/include -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/histctrl/swp -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/jr3dparty/opencv4 -FPIC -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/src/pymodule/. /usr/bin/gcc -C . /usr/bin/common/sample_common.Ldc -o . /usr/bin/common/sample_common.Ldc
```

```
aarch64-linux-gnu-gcc -DS5928V100 -DOR_XXX -lpthread -ln -ld -DISP V2 -lstdc++ -Wall -fsigned-char -mcpu=cortex-a53 -fno-aggressive-loop-optimizations -ld -ffunction-sections -fdats-sections -O2 -fstack-protector-strong -fpic -D_FILE_OFFSET_BITS=64 -DLARGEFILE_SOURCE -D_LARGEFILE64_SOURCE -PIE -pie -s -Wl,-flang=chdr -DOT_RELAX -OVER_XU1 -OVER_YuB -OVER_ZuB -OVER_B=10 -OVERS_BIT_64 -DKERNEL_BIT_64 -Wno-date-time -DENOSDR0_TYPE=SONY_IMX374_SLAVE_MIPI_A0_30FPS_12BIT -DENOSDR1_TYPE=SONY_IMX374_SLAVE_MIPI_A0_30FPS_12BIT -DENOSDR2_TYPE=HLSI_CS1860_MIPI_A0_30FPS_10BIT -DENOSDR3_TYPE=HLSI_CS1860_MIPI_A0_30FPS_10BIT -DOT_ACUTE_TYPE_HMT -DOT_ACUTE_TYPE_INNER -DOT_USE_STATIC_MODULE_REGISTER -DOT_MAC_USE_STATIC_MODULE_REGISTER -DOT_MAC_HAVE_SBR_LN -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/src/common -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/histctrl/mip -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/histctrl/inc/exp -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/histctrl/osal/include -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/histctrl/swp -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/include/jr3dparty/opencv4 -FPIC -I/home/eabina/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/src/pymodule/. /usr/bin/gcc -C . /usr/bin/common/sample_common.Ldc -o . /usr/bin/common/sample_common.Ldc
```

```
eabina@ubuntu22:~/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/src/pymodule$ ls
```

```
include      Makefile     sample_pymodule.cpp  sample_pymodule.so
```

```
eabina@ubuntu22:~/workspace/S5928V100_SDK_V2.0.2.2_MPP_Sample/src/pymodule$
```

(5) 板端验证

a. 先运行 vio 程序开启两个 VPSS 通道(通道深度为 1), VO 显示 4 分屏。

```

-----vpss grp video frame info-----
grp  width  height  pixel_fmt  video_format  dynamic_range  cmp_mode  first_pic_pts  cur_pic_pts
0      2592   1520    YUV-SP420    LINEAR        SDR8         NONE          0              0
-----vpss chn output status-----
grp  chn  enable  width  height  pixel_fmt  video_format  dynamic_range  cmp_mode  send_ok  frame_rate  proc_time
0      0      Y    2592   1520    YUV-SP420    LINEAR        SDR8         NONE      4390      60         7031
0      1      Y    640    640     YUV-SP420    LINEAR        SDR8         NONE       0         60         7185

```

b. python 获取 vpss 视频帧, 使用 opencv 保存图片:

```
import sys
import cv2
import numpy as np

import sample_pymodule

def test_sample_module():
    VPSS_GRP = 0
    VPSS_CHN = 1
    VO_LAYER = 0
    VO_CHN = 1

    frame = sample_pymodule.py_vpss_get_chn_frame(VPSS_GRP, VPSS_CHN)
    print("frame.shape =", frame.shape, "dtype =", frame.dtype)

    # do something for frame

    # 转灰度
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # 二值化 (阈值 128)
    _, binary = cv2.threshold(gray, 128, 255, cv2.THRESH_BINARY)

    # 转回三通道
    binary_bgr = cv2.cvtColor(binary, cv2.COLOR_GRAY2BGR)
```

```
cv2.imwrite("./output/output_frame.jpg", frame)
ret = sample_pymodule.py_vo_send_frame(VO_LAYER, VO_CHN, binary_bgr, VPSS_GRP, VPSS_CHN)

print("py_vo_send_frame ret =", ret)

sample_pymodule.py_vpss_release_chn_frame(VPSS_GRP, VPSS_CHN)

if __name__ == "__main__":
    print("py module test")
    test_sample_module()
```

```
root@localhost:~/pymodule# ls
output  qt_ive_vio  sample_pymodule.so  test_sample_pymodule.py  test_sample_pymodule_real_time.py
root@localhost:~/pymodule# python3 test_sample_pymodule.py
py module test
frame.shape = (640, 640, 3) dtype = uint8
py_vo_send_frame ret = 0
root@localhost:~/pymodule# ls output/
output_frame.jpg
root@localhost:~/pymodule#
```

c. python 获取 vpss 视频帧，opencv 处理送给 vo 实时显示:

```
import sys
import cv2
import numpy as np
import time

import sample_pymodule

def test_sample_module():
    print("开始循环读取并处理帧... 按 Ctrl+C 退出")

    VPSS_GRP = 0
    VPSS_CHN = 1
    VO_LAYER = 0
    VO_CHN = 1

    while True:
        try:
            # 获取一帧
            frame = sample_pymodule.py_vpss_get_chn_frame(VPSS_GRP, VPSS_CHN)
        except Exception as e:
            print("获取帧失败:", e)
            continue

        try:
            # do something for frame
            # 转灰度
```



```
gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

# 二值化 (阈值 128)
_, binary = cv2.threshold(gray, 128, 255, cv2.THRESH_BINARY)

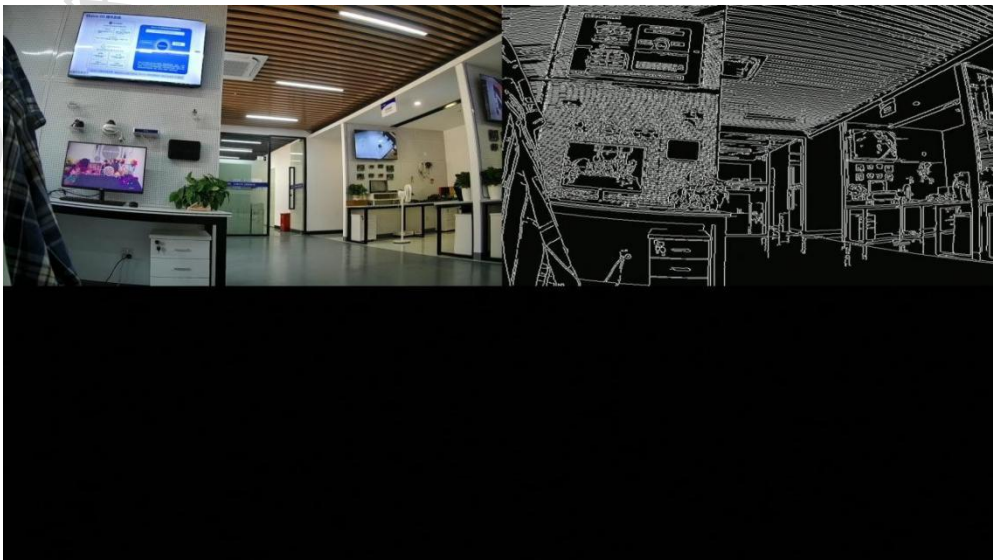
# 转回三通道
binary_bgr = cv2.cvtColor(binary, cv2.COLOR_GRAY2BGR)

try:
    sample_pymodule.py_vo_send_frame(VO_LAYER, VO_CHN, binary_bgr, VPSS_GRP, VPS
S_CHN)
except Exception as e:
    print("VO 输出失败:", e)

finally:
    # 无论成功与否都要释放
    try:
        sample_pymodule.py_vpss_release_chn_frame(VPSS_GRP, VPSS_CHN)
    except Exception as e:
        print("释放帧失败:", e)

if __name__ == "__main__":
    print("py module test")
    try:
        test_sample_module()
    except KeyboardInterrupt:
        print("\n 已手动中断")
```

```
root@localhost:~/pymodule# ls
output qt_ive_vio sample_pymodule.so test_sample_pymodule.py test_sample_pymodule_real_time.py
root@localhost:~/pymodule# python3 test_sample_pymodule_real_time.py
py module test
开始循环读取并处理帧... 按 Ctrl+C 退出
```





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